GLIDECART FINANCIAL PLAN

A Five Research Pillars.

KEY RISKS AND MITIGATIONS FOR THE GLIDECART

PROFESSORS: ZHANDRY, ANSHU, SCHOELKOPF, SHADBOLT, AND MONROE.

A BOARD-LEVEL SUMMARY of the key risks and mitigations for the GlideCart roadmap, aligned with the five research pillars (Zhandry, Anshu, Schoelkopf, Shadbolt, Monroe). A concise, decision-oriented, and suitable for a risk register slide or board memo

1. OPERATIONAL RELIABILITY RISK

Risk: Robots fail or behave inconsistently in noisy, crowded retail environments. Thereby, they are down-time, customer disruption, and higher support costs.

Primary Impact:

- Increased OpEx
- Store-level safety incidents
- Loss of retailer trust

MITIGATION (LEAD PILLAR):
Rob Schoelkopf — **Noise-Resilient Systems Guarantees**

- Noise-tolerant communication and sensing
- Stability-first optimization
- Proactive anomaly detection

Residual Risk Level: Low once fully implemented
Board View: Highest near-term risk; address first

2. CYBERSECURITY AND TRUST RISK

Risk: Payment fraud, robot impersonation, or data breaches: future quantum-enabled attackers.

Primary Impact:

- Regulatory exposure
- Brand damage
- Financial loss

MITIGATION (LEAD PILLAR):
Mark Zhandry — **Post-Quantum Cryptography**

- Quantum-resistant authenticate and signatures
- Anti-forgery and replay protection
- Secure payment and identity workflows

Residual Risk Level: LOW

Board View: Non-negotiable foundation; must precede scale

3. DECIDE INTEGRITY AND EFFICIENT RISK

Risk: Robots is incorrect or slow due to inefficient communication or adversarial inference.

Primary Impact:

- Poor customer experience
- Reduced throughput
- Higher cloud and network costs

MITIGATION (LEAD PILLAR):
Anshu — **Secure Communication and Complex Guarantees**

- Minimal, secure data exchange
- Decision logic resistant to inference
- Quantum secure random for authentication

Residual Risk Level: A Medium-Low as scale increases
Board View: Primary margin-expansion lever

4. SCALABILITY AND FLEET GROWTH RISK

Risk: Fleet performance degrades as robots are de-ployed; network costs grow linearly with scale.

Primary Impact:

- Margin erosion
- Delayed expansion
- Competitive disadvantage

MITIGATION (LEAD PILLAR):
Pete Shadbolt — **Photonic Quantum Communication**

- High-bandwidth, low-latency optical links
- Quantum-safe key distribution using light
- Secure fleet-level coordination

Residual Risk Level: LOW AT LARGE SCALE

Board View: GROW UNLOCK; GATED TO FLEET DENSITY

5. QUANTUM THREAT AND VALIDATION RISK

Risk: Secue assumptions fail against real quantum adversaries; vulnerabilities too late.
Primary Impact:

- Long-term security debt
- Forced redesigns
- Loss of “quantum-ready” credibility

MITIGATION (LEAD PILLAR):
Chris Monroe (IonQ) — **Trapped-Ion Simulation and Validation**

- High-fidelity quantum attack simulations
- Stress-test cryptography, routing, and optimization
- Certification-grade validation for regulators and partners

Residual Risk Level: Very Low
Board View: Low-cost insurance against long-term disruption

CONSOLIDATED RISK-TO-PILLAR MAPPING

Risk Category	Primary Mitigation	Secondary Support
Reliability and uptime	Schoelkopf	Anshu
Cybersecurity and payments	Zhandry	Monroe
Decision correctness and latency	Anshu	Schoelkopf
Fleet scalability	Shadbolt	Anshu
A quantum threats	Monroe	Zhandry, Shadbolt

Board-Level Conclusions

- **Largest immediate risk:** Operational reliability and mitigated by Schoelkopf
- **Largest existential risk:** Security failure and mitigated by Zhandry
- **Largest margin risk:** Inefficiency and mitigated by Anshu
- **Largest growth risk:** Fleet scaling and mitigated by Shadbolt
- **Largest long-term uncertainty:** Quantum disruption and mitigated by Monroe
- **Net effect:** The roadmap systematically reduces downside risk first, then enables profitable growth, while keeping future quantum threats contained.

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QUANTIFIED FINANCIAL DOWNSIDE PER RISK

(ANNUALIZED IMPACT AT MODERATE FLEET SCALE)

HOW TO READ THIS

- Ranges reflect direct costs + secondary effects (lost sales, support, churn).
- "Without mitigation" = exposure if the roadmap is not executed.
- "With mitigation" = residual exposure after full pillar deployment.

RISK-BY-RISK FINANCIAL EXPOSURE

RISK CATEGORY	PRIMARY COST DRIVERS	DOWNSIDE WITHOUT MITIGATION	RESIDUAL DOWNSIDE WITH MITIGATION	PRIMARY MITIGATION PILLAR
Operational Reliability (Downtime)	Lost sales, staff idle time, emergency fixes	$6–12M / year	$1–2M /year	Schoelkopf
Cybersecurity and Payments	Breach response, fines, churn, PR	$3–6M per incident	<$0.5M per incident	Zhandry and Monroe
Decision Errors / Inefficiency	Misrouting, slow checkout, cloud waste	$2–4M / year	$0.5–1M /year	Anshu
Fleet Scaling Bottlenecks	Extra networking, delayed expansion	$3–5M / year	$1–2M /year	Shadbolt
Future Quantum Threats	Forced redesign, emergency upgrades	$8–15M one-time	<$1M one-time	Monroe

EVIDENCE ANCHORS (INDUSTRY-LEVEL)

- **Robot/system downtime:** cost $1K–$10K per minute in automated environments, and retail outages reaching >$1M per hour in peak periods (formant.io, linkedin.com, 2026).
- **Retail data breaches:** average $3.48–$4.9M per incident, rising 18% YoY (retailtouchpoints.com, upwind.io, 2026)
- **Autonomous retail efficiency gaps:** delays in deployment lead to multi-million dollar preventable losses per chain annually (pipelinepub.com, 2026).

CUMULATIVE DOWNSIDE AT RISK (BOARD VIEW)

WITHOUT THE ROADMAP

- **Annual recurring exposure:** $11–21M
- **Tail-risk events (quantum / breach):** $8–15M+ one-time
- High volatility and unpredictable cash impact

WITH THE ROADMAP FULLY EXECUTED

- **Annual residual exposure:** $3–6M
- **Tail-risk events:** Contained and planned
- Predictable cost envelope
- 65–75% downside risk reduction

PILLAR-TO-RISK LEVERAGE	(WHY SEQUENCING MATTERS)
Pillar	% of Downside Risk Reduced
Schoelkopf	30–35%
Zhandry	20–25%
Anshu	15–20%
Shadbolt	10–15%
Monroe	10–15% (tail-risk insurance)

Board-Level Takeaway

The GlideCart roadmap converts an $11–21M annual downside into a manageable $3–6M operating risk while eliminating existential tail risks.

THIS IS WHY:

- Schoelkopf is funded first (largest recurring loss)
- Zhandry is non-negotiable (existential risk)
- Anshu protects margins
- Shadbolt protects growth
- Monroe prevents future shocks

GlideCart Risk Heatmap

EXECUTIVE VIEW: OVERALL RISK POSTURE

IMPROVE RISKS IDENTIFIED, SEQUENCED, AND ACTIVELY MITIGATED VIA TARGETED INVESTMENT.

RISK HEATMAP MATRIX

IMPACT ☐ LIKELIHOOD ☐ LOW ☐ MEDIUM ☐ HIGH

IMPACT \ LIKELIHOOD	LOW	MEDIUM	HIGH
High Impact	Future Quantum Threat	Fleet Scalability	Operational Reliability
Medium Impact	Optimization Efficiency	Cybersecurity & Trust	—
Low Impact	Regulatory Drift	—	—

RISK REGISTER SUMMARY (BOARD DETAIL)

RISK CATEGORY	IMPACT	LIKELIHOOD	TREND	PRIMARY PILLAR	MITIGATE STRENGTH
Operational Reliability	High	High-Medium	Improving	Schoelkopf	Strong
Cybersecurity & Trust	High	Medium-Low	Improving	Zhandry	Strong
Decide Integrity & Efficiency	Medium	Medium-Low	Improving	Anshu	Moderate-Strong
Fleet Scalability	High	Medium	Stable	Shadbolt	Emerging
Future Quantum Threats	High	Low (now)	Long-term	Monroe	Strong

RISK OWNERSHIP & MITIGATION MAPPING

RISK	WHAT GO WRONG	PRIMARY MITIGATION	SECONDARY SUPPORT
Reliability failure	Downtime, store incidents	Schoelkopf	Anshu
Security breach	Fraud, data loss	Zhandry	Monroe
Slow/ incorrect decisions	Poor CX, cost overruns	Anshu	Schoelkopf
Scaling bottlenecks	Margin erosion	Shadbolt	Anshu
Quantum attack surprise	Strategic disruption	Monroe	Zhandry, Shadbolt

RESIDUAL RISK AFTER FULL ROADMAP EXECUTION

Risk Category	Residual Risk
Operational Reliability	Low
Cybersecurity & Trust	Low
Decision Integrity	Low
Fleet Scalability	Medium-Low
Future Quantum Threats	Very Low

BOARD-LEVEL INTERPRETATION

Most urgent risk: Operational reliability. Addressed first via Schoelkopf.
Most existential risk: Security failure. Locked down via Zhandry + Monroe.
Most margin-sensitive risk: Inefficiency. Reduced via Anshu.
Most growth-limiting risk: Scaling bottlenecks. Addressed via Shadbolt.
Most long-term uncertainty: Quantum disruption. Contained via Monroe's validation layer.

EXECUTIVE TAKEAWAY

All critical GlideCart risks are either actively declining or strategically contained, with no single point of failure remaining unmitigated.

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12-Quarter Funding Roadmap
(DOLLAR RANGES)

Assumption: Quarterly platform + R&D budget grows from $4.0–$4.5M in Year 1 to $6.0–$6.5M by Year 3 as fleet scale increases.

YEAR 1 — RELIABILITY & SECURITY FIRST (Q1–Q4)

Quarterly Budget: $4.0–$4.5M

Quarter	SCHOELKOPF (RELIABILITY)	ZHANDRY (SECURITY)	ANSHU (EFFICIENCY)	SHADBOLT (SCALE)	TOTAL
Q1	$1.8–2.0M	$1.3–1.6M	$0.6–0.7M	$0.2M	$4.0–4.5M
Q2	$1.8–2.0M	$1.2–1.4M	$0.7–0.9M	$0.2M	$4.0–4.5M
Q3	$1.6–1.8M	$1.2–1.4M	$0.9–1.1M	$0.2M	$4.0–4.5M
Q4	$1.4–1.6M	$1.2–1.4M	$1.1–1.3M	$0.2M	$4.0–4.5M

Gate: ≥98% comms integrity under noise; zero security incidents.

YEAR 2 — EFFICIENCY & MARGIN EXPANSION (Q5–Q8)

Quarterly Budget: $5.0–$5.5M

QTR	SCHOELKOPF	ZHANDRY	ANSHU	SHADBOLT	TOTAL
Q5	$1.5–1.7M	$1.2–1.4M	$1.7–1.9M	$0.5M	$5.0–5.5M
Q6	$1.5–1.7M	$1.2–1.4M	$1.7–1.9M	$0.5M	$5.0–5.5M
Q7	$1.3–1.5M	$1.0–1.1M	$1.7–1.9M	$1.0–1.1M	$5.0–5.5M
Q8	$1.3–1.5M	$1.0–1.1M	$1.5–1.7M	$1.2–1.4M	$5.0–5.5M

Gate: 25–30% OpEx/robot reduction; successful photonic pilot.

Quarterly Budget: $6.0–$6.5M

QUARTER	SCHOELKOPF	ZHANDRY	ANSHU	SHADBOLT	TOTAL
Q9	$1.2–1.3M	$0.9–1.0M	$1.5–1.6M	$2.4–2.6M	$6.0–6.5M
Q10	$1.2–1.3M	$0.9–1.0M	$1.2–1.3M	$2.7–2.9M	$6.0–6.5M
Q11	$0.9–1.0M	$0.9–1.0M	$1.2–1.3M	$3.0–3.2M	$6.0–6.5M
Q12	$0.9–1.0M	$0.9–1.0M	$1.2–1.3M	$3.0–3.2M	$6.0–6.5M

Targets: 3–5× robots per region; 35–40% OpEx/robot reduction.

3-YEAR CUMULATIVE SPEND (RANGES)

PILLAR	12-QUARTER TOTAL
Schoelkopf (Reliability)	$16–18M
Zhandry (Security)	$11–13M
Anshu (Efficiency)	$13–15M

Board Takeaway

- Front-loaded reliability and security reduce downside risk early.
- Mid-cycle efficiency improves unit economics.
- Back-loaded photonics enables profitable scale without linear OpEx growth.

If you want, I can overlay these ranges onto the KPI gates, convert to a single CFO slide, or extend the table to include Chris Monroe (IonQ) as a validation/testing pillar alongside Shadbolt for quantum-attack simulation.

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12-QUARTER FUNDING ROADMAP

INTEGRATED FIVE-PILLAR INVESTMENT PLAN (WITH DOLLAR RANGES)

PILLARS

- Rob Schoelkopf — Reliability & real-world stability
- Mark Zhandry — Post-quantum security & trust
- Anshu — Efficiency & margin expansion
- Pete Shadbolt — Scalable photonic growth
- Chris Monroe (IonQ) — Quantum-attack simulation & validation

STRATEGIC ROLE OF CHRIS MONROE

Chris Monroe does not replace any pillar.

HE STRENGTHENS ALL OF THEM BY PROVIDING:

- High-fidelity trapped-ion simulations of quantum attacks
- Validation of cryptography (Zhandry) under realistic quantum conditions
- Testing decision-logic resilience (Anshu)
- Stress-testing fleet optimization models (Schoelkopf)
- Benchmarking future photonic security (Shadbolt)

Monroe's research is capital-efficient and insurance-like: small spend, high downside protection.

YEAR 1 — RELIABILITY, SECURITY, AND EARLY VALIDATION (Q1–Q4)

QUARTERLY BUDGET: $4.0–$4.5M

QUARTER	SCHOELKOPF	ZHANDRY	ANSHU	SHADBOLT	MONROE	TOTAL
Q1	$1.6–1.8M	$1.3–1.5M	$0.6–0.7M	$0.2M	$0.3M	$4.0–4.5M
Q2	$1.6–1.8M	$1.2–1.4M	$0.7–0.9M	$0.2M	$0.3M	$4.0–4.5M
Q3	$1.5–1.7M	$1.2–1.3M	$0.9–1.0M	$0.2M	$0.3M	$4.0–4.5M
Q4	$1.4–1.6M	$1.2–1.3M	$1.0–1.2M	$0.2M	$0.3M	$4.0–4.5M

Monroe focus (Y1):

- Quantum-attack simulations against authentication & routing
- Early validation of crypto and decision logic

YEAR 2 — EFFICIENCY, SCALING PILOTS, AND DEEP VALIDATION (Q5–Q8)

QUARTERLY BUDGET: $5.0–$5.5M

QUARTER	SCHOELKOPF	ZHANDRY	ANSHU	SHADBOLT	MONROE	TOTAL
Q5	$1.4–1.6M	$1.1–1.3M	$1.6–1.8M	$0.5M	$0.4M	$5.0–5.5M
Q6	$1.4–1.6M	$1.1–1.3M	$1.6–1.8M	$0.5M	$0.4M	$5.0–5.5M
Q7	$1.2–1.4M	$1.0–1.1M	$1.6–1.8M	$1.0–1.1M	$0.4M	$5.0–5.5M
Q8	$1.2–1.4M	$1.0–1.1M	$1.4–1.6M	$1.2–1.3M	$0.4M	$5.0–5.5M

Monroe focus (Y2):

- Simulate quantum adversaries at increasing scale
- Validate fleet-level optimization and communications
- Identify vulnerabilities before scale-out

YEAR 3 — SCALE, PHOTONICS, AND CONTINUOUS QUANTUM READINESS (Q9–Q12)

QUARTERLY BUDGET: $6.0–$6.5M

QUARTER	SCHOELKOPF	ZHANDRY	ANSHU	SHADBOLT	MONROE	TOTAL
Q9	$1.1–1.2M	$0.9–1.0M	$1.4–1.5M	$2.3–2.5M	$0.6–0.7M	$6.0–6.5M
Q10	$1.1–1.2M	$0.9–1.0M	$1.1–1.2M	$2.6–2.8M	$0.6–0.7M	$6.0–6.5M
Q11	$0.9–1.0M	$0.9–1.0M	$1.1–1.2M	$3.0–3.1M	$0.6–0.7M	$6.0–6.5M
Q12	$0.9–1.0M	$0.9–1.0M	$1.1–1.2M	$3.0–3.1M	$0.6–0.7M	$6.0–6.5M

MONROE FOCUS (Y3):

- Continuous quantum-threat simulation
- Certification-grade testing for customers and regulators
- Long-term quantum-readiness assurance

3-YEAR CUMULATIVE SPEND (UPDATED)

PILLAR	12-QUARTER TOTAL
Schoelkopf (Reliability)	$15–17M
Zhandry (Security)	$11–12M
Anshu (Efficiency)	$13–14M
Shadbolt (Scale)	$8–10M
Monroe (Validation)	$4–5M
Total (3 Years)	**$51–58M**

BOARD-LEVEL TAKEAWAY

- Schoelkopf ensures GlideCart works
- Zhandry ensures it's trusted
- Anshu ensures it's efficient
- Shadbolt ensures it scales
- Monroe ensures it stays safe against future quantum threats

Monroe adds asymmetric risk protection at modest cost — making him the most efficient "insurance policy" in the roadmap.

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EBITDA IMPACT BY RISK (EXECUTIVE VIEW)

ASSUMPTIONS (FOR COMPARABILITY)

- Moderate multi-store deployment
- EBITDA margins typical of early-scale retail automation platforms
- Impacts shown as annualized EBITDA effect
- Ranges reflect best estimate bands, not point forecasts

EBITDA IMPACT PER RISK CATEGORY

Risk Category: Mitigation	**Primary Pillar:** EBITDA Impact Without Mitigate	EBITDA Impact With Mitigate	EBITDA Uplift from	PROFESSOR
Operational Reliability (Downtime)	–$4 to –$8M / year	–$0.8 to –$1.5M / year	+$3 to +$6M / year	**Schoelkopf**
Cybersecurity & Payments	–$3 to –$6M per incident	–<$0.5M per incident	+$2.5 to +$5.5M (event-driven)	**Zhandry + Monroe**
Decision Errors/Inefficiency	–$1.5 to –$3M / year	–$0.4 to –$0.8M / year	+$1 to +$2.2M / year	**Anshu**
Fleet Scaling Bottlenecks	–$2 to –$4M / year	–$0.7 to –$1.5M / year	+$1.3 to +$2.5M / year	**Shadbolt**
Future Quantum Disruption	–$8 to –$15M one-time hit	–<$1M planned cost	+$7 to +$14M risk avoided	**Monroe**

EBITDA Impact Summary (Annualized)

Without the Roadmap

- **Recurring EBITDA drag:** –$7.5 to –$15M / year
- **Tail-risk events:** –$8 to –$15M one-time
- EBITDA volatility is high and unpredictable

With the Full Roadmap

- **Residual EBITDA drag:** –$2 to –$4M / year
- **Tail-risk events:** Contained and budgeted
- EBITDA becomes stable and forecastable

Net EBITDA improvement

+$5 to +$11M annually
+$7 to +$14M in avoided catastrophic downside

EBITDA LEVERAGE BY RESEARCH PILLAR

PILLAR	EBITDA CONTRIBUTION TYPE	RELATIVE EBITDA IMPACT
Schoelkopf	Recurring OpEx reduction	
Zhandry	Loss prevention (security)	
Anshu	Margin expansion	
Shadbolt	Growth-margin protection	
Monroe	Tail-risk insurance	

EBITDA TIMING PROFILE (BOARD INSIGHT)

- **Near-term (0–12 months):** Schoelkopf + Zhandry deliver immediate EBITDA protection
- **Mid-term (12–24 months):** Anshu improves margins per transaction
- **Long-term (24–36 months):** Shadbolt enables EBITDA-positive scale
- Monroe prevents future EBITDA shocks

BOARD-LEVEL CONCLUSION

The GlideCart roadmap is EBITDA-accretive, not just risk-reducing.

- Reliability improvements drive recurring EBITDA uplift
- Security investments prevent large negative EBITDA shocks
- Efficiency and scale convert growth into margin expansion
- Quantum validation avoids existential write-downs

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QUARTERLY EBITDA RISK DASHBOARD

QUARTER: ☐ Q1 ☐ Q2 ☐ Q3 ☐ Q4 YEAR: ☐ 2026
OVERALL EBITDA RISK STATUS: ☐ IMPROVING ☐ STABLE

1. EBITDA RISK OVERVIEW (AT-A-GLANCE)

RISK CATEGORY	EBITDA EXPOSURE (UNMITIGATED)	EBITDA AT RISK (CURRENT)	TREND	OWNER
Operational Reliability	–$4 to –$8M / yr	–$1.0 to –$1.5M	☐	Schoelkopf
Cybersecurity & Payments	–$3 to –$6M / event	–<$0.5M / event	☐	Zhandry / Monroe
Decision Inefficiency	–$1.5 to –$3M / yr	–$0.4 to –$0.8M	☐	Anshu
Fleet Scaling Bottlenecks	–$2 to –$4M / yr	–$0.7 to –$1.5M	☐	Shadbolt
Future Quantum Disruption	–$8 to –$15M (one-time)	–<$1M (planned)	☐	Monroe
Total EBITDA at Risk (Quarterized): ☐ $0.6–$1.2M				
Total EBITDA Avoided (vs. baseline): ☐ $1.5–$3.0M / quarter				

2. KEY RISK METRICS (QUARTERLY KPIS)

RISK	KPI	TARGET	CURRENT	STATUS
Reliability	Robot uptime	>98%	☐ %	☐/☐/☐
Reliability	Unplanned downtime	☐ QoQ	☐ hrs	☐/☐/☐
Security	Incidents	0	☐	☐/☐/☐
Decisions	Latency	–30% YoY	☐ ms	☐/☐/☐
Scale	Robots/store supported	☐ QoQ	☐	☐/☐/☐
Quantum	Attack simulations run	≥ planned	☐	☐/☐/☐

3. EBITDA IMPACT BREAKDOWN (QUARTER)

PILLAR	EBITDA IMPROVEMENT THIS QUARTER	SOURCE OF IMPROVEMENT
Schoelkopf	+$0.7–$1.2M	Less downtime, fewer support calls
Zhandry	+$0.3–$0.6M	Avoided fraud & incident response
Anshu	+$0.2–$0.4M	Lower cloud/network cost
Shadbolt	+$0.1–$0.3M	Improved fleet utilization
Monroe	Risk avoided	Quantum-attack validation
Net EBITDA Uplift (Quarter): +$1.3–$2.5M		

4. EBITDA IMPACT BREAKDOWN (QUARTER)

RISK	LAST QUARTER	THIS QUARTER	DIRECTION
Reliability	☐	☐	☐ Improving
Security	☐	☐	☐ Improving

Decision Efficiency	☐	☐	☐ Improving
Scalability	☐	☐	☐ Stable
Quantum Threat	☐	☐	☐ Improving

5. MANAGEMENT ACTIONS (NEXT QUARTER)

☐ Increase reliability spend if uptime <98%

☐ Advance Shadbolt photonic milestones if fleet density threshold met

☐ Expand Monroe simulations if new crypto or routing deployed

☐ Freeze scale-out if security KPI deviates from target

Board-Level Interpretation

- **Largest EBITDA driver:** Operational reliability (Schoelkopf)
- **Largest tail-risk reducer:** Cybersecurity + quantum validation (Zhandry + Monroe)
- **Margin lever:** Decision efficiency (Anshu)
- **Growth enabler:** Fleet scalability (Shadbolt)

EBITDA volatility is declining quarter-over-quarter as risk moves from reactive to managed.

One-Line Board Summary

GlideCart's quarterly EBITDA risk is controlled, declining, and increasingly predictable—with no unmitigated high-impact risks remaining.

GlideCart QUARTERLY EBITDA RISK DASHBOARD

WITH RED / YELLOW / GREEN THRESHOLDS

QUARTER: ☐ Q1 ☐ Q2 ☐ Q3 ☐ Q4 **YEAR:** ☐ 2026

1. EBITDA RISK STATUS (THRESHOLD-BASED)

RISK CATEGORY	GREEN (ON TRACK)	YELLOW (WATCH)	RED (INTERVENE)	OWNER
Operational Reliability	EBITDA drag ≤ $0.5M/qtr	$0.5–$1.0M/qtr	> $1.0M/qtr	Schoelkopf
Cybersecurity & Payments	0 incidents	Near-misses only	≥1 incident	Zhandry / Monroe
Decision Inefficiency	≤ $0.25M/qtr	$0.25–$0.5M/qtr	> $0.5M/qtr	Anshu
Fleet Scalability	Cost/task flat or ☐	Cost/task ☐ ≤5%	Cost/task ☐ >5%	Shadbolt
Future Quantum Threat	Simulations passing	Gaps identified	Unmodeled exposure	Monroe

2. KPI THRESHOLDS (WHAT TURNS RED)

OPERATIONAL RELIABILITY

KPI	GREEN	YELLOW	RED
Robot uptime	≥98%	95–97.9%	<95%
Unplanned downtime	≤1.5 hrs/robot/qtr	1.5–3 hrs	>3 hrs
Support tickets	☐ QoQ	Flat QoQ	☐ QoQ

3. CYBERSECURITY & PAYMENTS

PI	GREEN	YELLOW	RED
Security incidents	0	Attempted only	≥1 confirmed
Payment auth success	≥99.99%	99.9–99.99%	<99.9%
Mean detect time	<24 hrs	24–72 hrs	>72 hrs

4. DECISION INTEGRITY & EFFICIENCY

KPI	GREEN	YELLOW	RED
Decision latency	–30% YoY	–10–30%	<–10%
Routing errors	≤0.1%	0.1–0.3%	>0.3%
Cloud cost/txn	☐ ≥15% YoY	Flat	☐ YoY

5. FLEET SCALABILITY

KPI	Green	Yellow	Red
Robots/store supported	☐ QoQ	Flat QoQ	☐ QoQ
Fleet latency	☐ QoQ	Flat	☐ QoQ

Network cost/robot ☐ ≥20% YoY ☐ <20% Flat/☐

6. FUTURE QUANTUM READINESS

KPI	Green	Yellow	Red
Monroe simulations run	100% planned	75–99%	<75%
Critical gaps	0 open	≤2 tracked	>2 open
Certification readiness	On schedule	Minor delay	Off-track

3 EBITDA IMPACT BY PILLAR (QUARTER)

Pillar	Green EBITDA Gain	Yellow	Red
Schoelkopf	+$0.8–1.2M	+$0.3–0.7M	<+$0.3M

GLIDECART — QUARTERLY EBITDA RISK DASHBOARD

RED / YELLOW / GREEN EXECUTIVE VIEW.
OVERALL STATUS: IMPROVING & CONTROLLED
QUARTERLY EBITDA RISK: WITHIN TOLERANCE.

TREND: DECLINING RISK, INCREASING PREDICTABILITY

1. RISK HEATMAP (VISUAL CORE OF THE SLIDE)

RISK AREA	STATUS	EBITDA IMPACT (QUARTER)	TREND	OWNER
Operational Reliability	Green	–$0.8M to –$1.0M		**Schoelkopf**
Cybersecurity & Payments	Green	<$0.5M (event-driven)		**Zhandry / Monroe**
Decision Efficiency	Green	–$0.3M to –$0.4M		**Anshu**
Fleet Scalability	Yellow	–$0.7M to –$1.2M		**Shadbolt**
Future Quantum Threats	Green	<$0.25M (planned)		**Monroe**

LEGEND:

= Within EBITDA tolerance
= Monitor closely
= Immediate board action required

2. THRESHOLD INDICATORS

OPERATIONAL RELIABILITY (SCHOELKOPF)

Green: Uptime ≥ 98%, EBITDA drag ≤ $0.5M/qtr
Yellow: Uptime 95–97.9%\
Red: Uptime < 95% or EBITDA drag > $1.0M/qtr

CYBERSECURITY & PAYMENTS (ZHANDRY + MONROE)

Green: Zero incidents
Yellow: Attempted attacks only
Red: ≥1 confirmed breach

DECISION EFFICIENCY (ANSHU)

Green: Cloud cost/txn ≥15% YoY
Yellow: Flat QoQ
Red: Cost/txn or routing errors >0.3%

FLEET SCALABILITY (SHADBOLT)

Green: Cost per task or flat
Yellow: Cost per task ≤5%
Red: Cost per task >5%

QUANTUM READINESS (MONROE)

Green: 100% planned simulations passed
Yellow: Minor gaps tracked
Red: Unmodeled quantum exposure

3. EBITDA IMPACT BAR (VISUAL STRIP)

Reliability		+$0.8–1.2M
Security		+$0.3–0.6M
Efficiency		+$0.2–0.4M
Scalability		+$0.1–0.3M
Quantum Risk		Tail-risk avoided

Net EBITDA Uplift This Quarter: +$1.3–2.5M

4. MANAGEMENT ACTION TRIGGERS (AUTO-RESPONSE)

- **Any Red** Board notified within 24 hours
- **Reliability Red** Freeze expansion, reallocate +10% budget
- **Security Red** Incident protocol + regulator review
- **Scalability Yellow** Gate expansion, accelerate photonics

5. ONE-LINE BOARD TAKEAWAY

EBITDA risk is controlled, declining, and no high-impact risk is currently unmitigated.

GLIDECART - R/Y/G DASHBOARD GRAPHIC SUITABLE FOR A BOARD SLIDE.

VISUAL DASHBOARD (READY TO DROP INTO SLIDES)

HOW TO USE THIS SLIDE (BOARD CONTEXT)

- **Green bars** = within EBITDA tolerance, no action required
- **Yellow bar** = monitor closely, management attention required
- **Red (none currently)** = immediate board intervention

CURRENT STATUS INTERPRETATION

- **Operational Reliability** — under control (Schoelkopf impact visible)
- **Cybersecurity & Payments** — no incidents (Zhandry + Monroe)
- **Decision Efficiency** — margins improving (Anshu)
- **Fleet Scalability** — growth pressure emerging (Shadbolt gating point)
- **Future Quantum Threats** — validated and contained (Monroe)

RECOMMENDED SLIDE CAPTION

EBITDA risk is controlled and declining; only scaling remains a monitored growth risk.

A TRUE COLOR-CODED R/Y/G DASHBOARD WITH EBITDA $ IMPACT EMBEDDED INSIDE EACH BAR

READY FOR BOARD OR INVESTOR SLIDES

What This Slide Communicates Instantly

Each bar now conveys four executive signals at once:
1. Color (Red / Yellow / Green) current risk status
2. EBITDA $ impact quarterly financial effect
3. Trend arrow direction of risk movement
4. Relative magnitude where management attention matters most

Current Quarter Snapshot

- **Operational Reliability:** +$0.8–1.2M EBITDA (largest recurring driver)
- **Cybersecurity & Payments:** +$0.3–0.6M EBITDA (loss prevention)
- **Decision Efficiency:** +$0.2–0.4M EBITDA (margin expansion)
- **Fleet Scalability:** +$0.1–0.3M EBITDA (monitor as growth scales)
- **Future Quantum Threats:** Tail risk avoided (protects against $8–15M shocks)

Net quarterly EBITDA benefit: ≈ +$1.3–2.5M, with declining volatility.

Suggested Board Caption (Optional)

"EBITDA risk is controlled and improving; reliability and security are now EBITDA contributors, not cost centers."

Here is the updated visual R/Y/G dashboard with a clear last-quarter vs this-quarter EBITDA comparison built directly into each bar, ready for board review:

How the Board Should Read This Slide

Each horizontal bar is split into two halves:

- Left half: Last Quarter EBITDA impact
- Right half: This Quarter EBITDA impact
- Arrow: Direction of risk trend
- Color: Overall risk status (R / Y / G)

Current Quarter-over-Quarter Story

- Operational Reliability

+$0.5M +$1.0M EBITDA

Strong improvement from Schoelkopf-led reliability work

- Cybersecurity & Payments

+$0.2M +$0.4M EBITDA

Risk reduction accelerating (Zhandry + Monroe)

- Decision Efficiency

+$0.1M +$0.3M EBITDA

Margin expansion underway (Anshu)

- Fleet Scalability

+$0.1M +$0.2M EBITDA

Improving but still gated (Shadbolt)

- Future Quantum Threats

Risk Avoided

Tail risk now contained (Monroe validation)

Net effect: EBITDA improvement is broad-based and acwcelerating, with no category deteriorating.

Suggested Board Caption

"Quarter-over-quarter EBITDA risk is declining across all categories, with reliability and security delivering the largest gains."

What I Can Add Next (Instantly)

- Replace example values with actuals from finance
- Add next-quarter forecast markers
- Export as PowerPoint (16:9) or PDF
- Add executive owner initials inside each bar

Just tell me what you want layered in.

Glidecart R/Y/G dashboard with trend arrows added, ready for direct use in a board deck:

How to Read This Slide (Board-Ready)

COLOR = CURRENT RISK LEVEL

- **Green** — Within EBITDA tolerance
- **Yellow** — Monitor closely
- **Red** — Immediate board action required

Arrow = Direction of Risk

- **Improving** — Risk and EBITDA exposure declining
- **Stable** — Risk contained but not yet improving
- **Worsening** — Risk increasing (none currently)

Current Executive Interpretation

- **Operational Reliability** — improving due to Schoelkopf-led stability work
- **Cybersecurity & Payments** — improving with Zhandry + Monroe protections
- **Decision Efficiency** — improving via Anshu-driven optimization
- **Fleet Scalability** — stable, gated pending Shadbolt milestones
- **Future Quantum Threats** — improving, validated through Monroe simulations

Overall takeaway:

EBITDA risk is trending downward across all major categories,
with scaling risk appropriately gated and visible.

> “All high-impact risks are green or improving; scaling risk remains controlled and visible.”